

EBU6505 – Reasoning and Agents

Module Introduction

Dr Riasat Islam

Queen Mary University of London

Semester 1, 2026–2027

Teaching Team

Dr Riasat Islam

Lecturer in Computer Science
Programme Director, Intelligent Science and Technology

Teaching: Block 1 (Teaching Week 3) and Block 2 (Teaching Week 4)

Dr Chao Shu

Lecturer in Telecommunication Engineering
Programme Director, Telecommunications Engineering with Management

Teaching: Block 3 (Teaching Week 9) and Block 4 (Teaching Week 10)

We will connect formal ideas to practical systems, current AI tools, and responsible decision-making.

Meet Your Block 1 Lecturer

Dr Riasat Islam

I joined Queen Mary University of London as a Lecturer in August 2024.

- **BSc Engg in Electrical and Electronic Engineering**, Islamic University of Technology, Bangladesh (2009–2012)
- **MSc in ICT Innovation**, University College London and KTH Royal Institute of Technology (2013–2015)
- **PhD in Computing**, The Open University (2017–2023)
- Interests spanning artificial intelligence, human–computer interaction, healthcare, well-being, and education

<https://riasatislam.com>

Why This Module Now?

AI is rapidly changing

We hope this module helps you understand the underlying concepts and equips you with the skills to navigate this exciting era of AI.

- Move beyond using AI tools to understanding how reasoning systems work.
- Learn to question assumptions, inspect evidence, and evaluate outputs.
- Connect algorithms to real systems, people, organisations, and consequences.

Let's have fun!

What You Will Learn

- Model problems using states, actions, knowledge, uncertainty, and goals.
- Apply reasoning methods to solve problems and explain decisions.
- Build and critique knowledge-grounded and probabilistic agents.
- Use modern AI tools thoughtfully, with verification and reflection.
- Design systems that are useful, transparent, robust, and responsible.

The aim is not only to get an answer, but to understand why an answer should be trusted.

Block Overview

Block	Teaching week	Main themes
1	3	Reinforcement learning; knowledge representation; knowledge graphs; modern coding agents
2	4	Bayesian networks; exact inference; probabilistic reasoning over time; planning under uncertainty; DDNs and POMDPs
3	9	Further agent architectures and reasoning topics led by Dr Chao Shu
4	10	Further decision-making under uncertainty topics led by Dr Chao Shu

Blocks 1 and 2 establish the reasoning foundations; Blocks 3 and 4 extend them to wider agent and decision-making systems.

Teaching Schedule

Teaching week	Block	Lecturer
3	Block 1	Dr Riasat Islam
4	Block 2	Dr Riasat Islam
7–8	Coursework Mini Project 1	20%
9	Block 3	Dr Chao Shu
10	Block 4	Dr Chao Shu
11–12	Coursework Mini Project 2	20%
After week 16	Exam (date to be confirmed)	60%

How We Will Learn

Problem-solving based learning

We will work through concrete problems, compare possible approaches, explain our reasoning, and learn from mistakes.

AI-assisted learning: support, not substitution

AI can support your learning, but it does not replace your own reasoning. Start by thinking through the problem yourself, then use AI as a tutor, critic, or debugging partner.

Useful learning activities

- Ask for hints, explanations, alternatives, or counterexamples.
- Ask the tool to challenge assumptions, test edge cases, and debug code.

Your responsibility

- Verify outputs against lectures and reliable references.
- Explain every step in your own words; AI can be fluent, incomplete, or wrong.
- Follow each assessment brief: never outsource assessed reasoning or present generated work as your own.

Students who attend regularly, attempt the problems, and participate actively tend to perform very well.

Coursework – 40%

- Mini Project 1: 20%
Teaching Weeks 7–8
- Mini Project 2: 20%
Teaching Weeks 11–12

Exam – 60%

Held after Teaching Week 16.

Date and further arrangements will be confirmed later.

Coursework rewards sustained problem-solving across the module; the exam assesses your understanding and application of the core ideas.

- Collect feedback from classmates about the lectures.
- Assist lecturers in disseminating important information.
- Attend one Student–Staff Liaison Committee (SSLC) meeting during the semester.
- **Volunteers (one or two) will be welcomed or chosen during the first teaching week of the module.**

Module representatives help us listen, communicate, and improve the module together.